Technical Datasheet — VBF-T3R1-DSH-001

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Product

Item	Value	Source
Cell designation	V12_BalancedCooling (power-tool high-rate cell)	cell/p_v12.json
Chemistry	NMC811 / graphite, LiPF6 EC/EMC electrolyte	Chen2020 base
Nominal capacity	3.08 Ah (measured 1C: 3.075 Ah)	p_v12 / r4_v12_1c.json
Nominal energy	11.09 Wh	calc-energy
Cell mass	29.26 g (electrolyte excluded by contract)	calc-energy
Dimensions (active stack)	0.065 m x 1.58 m x 132 um	Chen2020 / layer sum
Voltage window	2.5 - 4.2 V	Chen2020

Rated performance (DFN-verified)

Specification	Rated	Measured	Margin	Source
Capacity @ 1C, 25 C	>= 2.0 Ah	3.075 Ah	+1.075 Ah	r4_v12_1c.json
5C discharge retention (5C cap / 1C cap)	>= 95 %	97.53 %	+2.53 pp	r4_v12_derived.json
Max temperature, 4C charge @ 45 C amb	<= 60 C	56.62 C	3.38 C below red line	r4_v12_4c.json
Plating at 4C charge	none	none (anode potential min +12.1 mV vs 0 V)	12.1 mV	r4_v12_4c.json
Power density	>= 4000 W/kg	32630 W/kg	x8.2	calc-energy

Additional characteristics

Item	Value	Source
Energy density	379.0 Wh/kg (gravimetric), 818.2 Wh/L (volumetric)	calc-energy
DC resistance (10% discharge)	4.41 mOhm	calc-energy
Discharge midpoint voltage	3.844 V	calc-energy
5C discharge capacity	2.999 Ah	r4_v12_5c.json
4C CC charge capacity (to 4.2 V)	0.808 Ah (CV phase required for full charge)	r4_v12_4c.json

Recommended operating conditions

- Continuous discharge: up to 5C (15.4 A); Fast charge: 4C CC (12.32 A) with CV taper, ambient <= 45 C.

- Cooling: heat transfer >= 20 W/m2/K (tool-body conduction + light airflow) — required to keep T <= 60 C.

- Charge termination 4.2 V; discharge cut-off 2.5 V.